

LAVAL-ENTRAMMES AP, France

WMO: 071340

Lat: 48.0306N

Lon: 0.7464W

Elev: 318

StdP: 14.53

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.8	27.7	19.0	14.5	28.4	22.4	17.1	29.3	25.3	52.3	22.8	51.2	5.7	10	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	87.0	68.3	82.8	66.6	79.2	65.4	70.2	82.7	68.2	79.2	66.6	76.4	7.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.9	96.8	74.2	64.4	91.5	72.4	62.9	86.7	70.9	34.3	82.7	32.6	79.3	31.4	76.6	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.7	18.1	16.0		20.9	93.4	4.0	3.7	18.0	96.1	15.6	98.3	13.4	100.4	10.5	103.1	
				20.1	73.9	4.0	2.4	17.2	75.6	14.8	77.0	12.6	78.3	9.6	80.1	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	53.2	41.7	42.0	46.4	51.1	56.4	63.1	66.2	65.4	61.1	54.9	47.3	42.1	(6)
(7)		DBStd	10.65	6.60	6.31	5.71	5.55	5.49	5.88	5.35	5.19	5.22	5.82	5.69	6.52	(7)
(8)		HDD50	1079	263	229	140	49	7	1	0	0	1	22	115	252	(8)
(9)		HDD65	4530	723	645	576	416	273	104	48	55	138	313	530	709	(9)
(10)		CDD50	2255	5	4	29	83	206	393	503	479	335	176	34	8	(10)
(11)		CDD65	232	0	0	0	0	6	47	87	69	22	1	0	0	(11)
(12)		CDH74	2254	0	0	0	14	74	494	831	630	203	8	0	0	(12)
(13)		CDH80	676	0	0	0	0	9	149	268	210	40	0	0	0	(13)
(14)	Wind	WSAvg	7.8	8.5	8.9	8.8	7.7	7.7	7.4	7.7	7.1	6.9	7.1	7.8	8.4	(14)
(15)	Precipitation	PrecAvg	31.5	3.1	2.6	2.2	2.3	2.6	2.0	2.2	2.0	2.2	3.3	3.2	3.5	(15)
(16)		PrecMax	39.8	5.4	5.3	6.9	5.0	4.8	4.1	5.6	5.2	6.3	7.9	5.9	7.6	(16)
(17)		PrecMin	23.1	1.1	0.4	0.4	0.4	0.6	0.2	0.5	0.2	0.5	1.0	1.0	0.9	(17)
(18)		PrecStd	5.2	1.3	1.4	1.5	1.3	1.0	1.1	1.3	1.4	1.5	1.7	1.4	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.3	59.2	68.4	76.0	80.7	91.8	92.3	92.7	84.6	74.4	62.7	57.3	(19)
MCWB			53.9	51.5	55.3	60.7	65.7	72.0	69.9	70.8	65.2	63.4	58.3	54.7	(20)	
2%		DB	54.6	55.3	63.0	71.1	75.4	84.0	86.8	86.1	79.8	69.3	60.0	55.5	(21)	
		MCWB	52.7	50.7	53.0	59.2	63.4	67.3	68.0	68.3	64.2	61.8	57.0	53.5	(22)	
5%		DB	53.0	53.0	59.3	66.5	71.2	79.5	82.5	80.8	75.4	66.1	57.9	53.8	(23)	
		MCWB	51.2	49.8	51.9	56.5	60.9	66.1	66.4	66.1	63.3	60.0	55.2	52.0	(24)	
10%		DB	51.0	51.1	55.8	62.1	67.4	75.4	78.5	76.5	71.6	63.4	55.9	52.0	(25)	
		MCWB	49.1	48.4	50.4	53.5	58.6	63.8	65.2	64.1	61.4	58.9	53.3	50.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.6	53.7	56.8	63.2	67.8	73.9	72.0	73.2	68.9	65.6	60.2	55.7	(27)
MCD B			55.6	56.1	65.0	72.2	77.8	86.1	88.3	87.4	78.4	71.3	61.4	56.6	(28)	
2%		WB	53.1	51.9	54.5	59.6	64.5	69.4	69.6	69.9	66.5	63.0	57.4	53.9	(29)	
		MCD B	54.3	54.2	60.0	69.5	73.6	80.2	82.4	83.0	74.9	67.1	59.4	55.0	(30)	
5%		WB	51.3	50.4	52.9	56.8	61.9	67.1	67.7	67.5	64.6	61.4	55.5	52.2	(31)	
		MCD B	52.6	52.6	57.5	64.7	69.8	77.4	79.1	77.5	72.9	64.8	57.5	53.6	(32)	
10%		WB	49.4	48.6	51.3	54.4	59.6	65.0	66.1	65.6	62.8	59.9	53.6	50.4	(33)	
		MCD B	50.8	50.8	55.4	60.9	65.9	73.6	76.4	73.9	69.7	62.9	55.6	51.9	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.6	11.9	15.0	17.7	17.5	18.9	19.3	19.0	18.7	14.2	11.0	9.9	(35)
MCD BR			10.1	14.3	20.6	24.3	23.5	25.9	27.2	26.7	24.3	16.9	12.1	9.9	(36)	
MCWBR			9.0	10.6	12.6	13.6	12.7	12.4	11.1	11.1	11.0	10.2	9.2	8.8	(37)	
5% WB		MCD BR	9.6	12.7	17.4	22.7	21.5	24.1	23.7	22.8	20.1	14.4	10.9	10.0	(38)	
		MCWBR	9.1	10.1	11.5	13.2	12.3	12.8	10.8	10.7	10.4	9.6	9.0	9.4	(39)	
(41)	Clear-Sky Solar Irradiance	taub	0.325	0.341	0.371	0.392	0.397	0.401	0.396	0.385	0.365	0.358	0.333	0.318	(41)	
taud		2.426	2.383	2.294	2.238	2.271	2.295	2.309	2.360	2.394	2.417	2.430	2.400	(42)		
Ebn at Noon		236	258	266	272	275	273	273	271	266	248	230	221	(43)		
Edh at Noon		21	27	35	41	41	40	39	36	31	26	21	19	(44)		
(45)	All-Sky Solar Radiation	RadAvg	324	588	937	1396	1650	1833	1749	1531	1209	697	400	278	(45)	
		RadStd	31	68	108	159	137	186	114	132	93	54	39	34	(46)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page